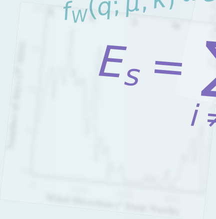
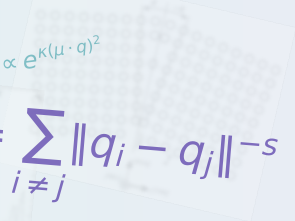
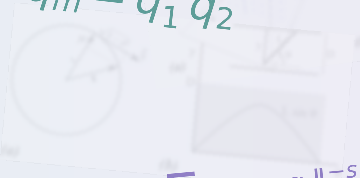
$$\bar{\mathbf{x}} = \frac{1}{n}\sum_i \mathbf{x}_i$$

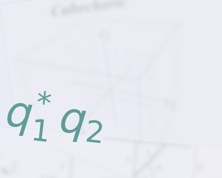




$$f_W(q;\mu,\kappa) \propto e^{\kappa(\mu\cdot q)^2}$$

$$E_s = \sum_{i\neq j} \|q_i - q_j\|^{-s}$$





$$q_m = q_1^* q_2$$

$$E_s = \sum_{i\neq j} \|q_i - q_j\|^{-s}$$

$$q_m$$

$$\kappa$$

$$\bar{\mathbf{x}} = \frac{1}{n}\sum_i \mathbf{x}_i$$

$$f_W(q;\mu,\kappa) \propto e^{\kappa(\mu\cdot q)^2}$$

$$\bar{C} = \frac{1}{n}\sum_i \cos\theta_i,\ \ \bar{S} = \frac{1}{n}\sum_i \sin\theta_i$$

$$g(\theta;\mu,\kappa) = \frac{1}{2\pi I_0(\kappa)}\, e^{\kappa\cos(\theta-\mu)}$$

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$$\theta \qquad \mu$$

$$q_m = q_1^* q_2$$

$$f(\mathbf{x};\mu,\kappa) \propto e^{\kappa\mu\cdot\mathbf{x}}$$

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